VERSA LOAN MANAGEMENT SYSTEM

Loan Management Module – Full Feature Breakdown

Customer Module – Loan Management

Document Information	
Version	2.0
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Category	Customer Module – Loan Management
System	Versa Loan Management System (VLMS)

Versa Loan Management System (LMS) - Features Breakdown

Process Flow (Key User Journeys)

1. Borrower Applies for a Loan

1. Sign up / Log in.
2. Select loan product (e.g., export credit).
3. Fill application form (auto-populate KYC details if returning user).
4. Upload documents (ID, financials).
5. Submit application → System generates reference number.
6. Receive email/SMS alerts on approval/rejection.

2. Loan Officer Processes Application

7. Review application (check documents, collateral).
8. Approve/reject (escalate to senior officer if amount exceeds limit).
9. If approved, disburse loan via chosen method.(If user has the permissions)
10. System updates loan account and notifies borrower.

3. Borrower Repays Loan

11. Log in to portal or visit agent.
12. Make repayment (bank transfer, mobile money).
13. System updates balance and sends receipt.
14. If late, auto-apply penalties and send reminders.

Process flow diagram

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4. Admin Monitors Portfolio

15. Access Dashboard

15.1. Log in to admin portal with elevated privileges.

15.2. System displays real-time KPIs:

I. Total disbursements (daily/weekly/monthly)

II. Portfolio-at-risk.

III. Arrears rate (%) and delinquency trends

IV. Expected Credit Loss (ECL) projections

16. Generate Reports

16.1. Automated Reports:

I. System generates scheduled reports (e.g., end-of-day delinquency list).

II. Export formats: PDF, Excel, or dashboard widgets.

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16.2. Ad-Hoc Reports:

I. Filter by: loan product, region, officer, or time period.

II. Advanced analytics (e.g., borrower risk scoring, collateral coverage).

17. Risk Monitoring & Alerts

17.1. System flags high-risk loans (auto-triggered rules):

I. Missed payments ≥ 2 installments.

II. Collateral value depreciation.

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17.2. Admin receives push notifications/SMS for critical alerts.

18. Recovery Actions

- Soft Recovery:
- System auto-sends reminders (email/SMS) to defaulters.
- Assign cases to recovery agents via workflow.
- Hard Recovery:
- Initiate legal process (system generates demand letters).
- Update borrower's credit status (blacklist if unresolved).

19. Audit & Compliance

- System logs all admin actions (view/edit) for audit trails.
- Generate regulatory reports (e.g., Central Bank requirements).

20. Portfolio Adjustments

- Admin can:
- Restructure delinquent loans (extend tenure/reduce
> installments).
- Write off irrecoverable loans (with senior officer
> approval).

Loan Product Management

Required Features

21. Loan Product Creation

a. Admin can define loan products with the following configurable parameters:

i. Product Types: Working capital, export credit, project financing, guarantee facilities, equity participation, hybrid facilities, project preparation.

ii. Financial Parameters:

21.1. Minimum and maximum loan amounts

21.2. Interest rate (fixed or variable)

21.3. Interest calculation methods (flat rate, reducing balance, annuity, linear amortization)

21.4. Repayment frequency (daily, weekly, monthly, quarterly, bullet payment)

21.5. Grace period (if applicable)

iii. Fees & Charges:

21.6. Processing fees (fixed or percentage-based)

21.7. Administration fees

21.8. Late payment penalties

21.9. Early repayment fees (if applicable)

21.10. Insurance fees (linked to collateral)

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21.11. Disbursement Options: Cash, cheque, bank transfer, or supply of materials.

21.12. Repayment Methods: Auto-debit, manual payment, mobile money, bank transfer.

21.13. Currency Support: Multi-currency (USD, MWK, ZAR, etc.) with auto-conversion for reporting.

22. Loan Product Modification

- Admin can update loan product details (e.g., interest rates, fees).
- Audit Trail:
- System logs all changes (who modified, what was changed, timestamp).
- Requires approval for critical changes (e.g., interest rate adjustments).
- Version Control:
- Existing loans retain original terms; changes apply only to new applications.

23. Product Approval Workflow

- New/updated products require approval from senior management.
- System sends notifications to approvers and tracks pending requests.

24. Loan Product Templates

- Predefined templates for common products (e.g., \"Agri-Loan Template\" with seasonal repayment schedules).
- Clone existing products to speed up setup.

25. Product Categories & Tagging
    - Categorize products by:
    - Duration (short-term, long-term).
    - Sector (agriculture, manufacturing, SMEs).
    - Risk level (low, medium, high).
    - Tags for easy filtering (e.g., \"collateral-free,\" \"emergency loan\").
26. Dynamic Pricing Rules
    - Risk-based pricing: Adjust interest rates based on:
    - Borrower's credit score.
    - Collateral value.
    - Loan amount/tenure.
    - Promotional rates (limited-time offers).
27. Multi-Currency Management
    - Admin can:
    - Set exchange rate sources (central bank, real-time APIs).
    - Define currency conversion rules for disbursements/repayments.
    - System auto-converts amounts for reporting in base currency.
28. Product-Level Permissions
    - Restrict product access by:
    - User roles (e.g., only senior officers can approve high-risk loans).

- Branch/region (e.g., rural branches offer only agri-loans).

29. Compliance Checks

- Enforce regulatory limits (e.g., maximum interest rate caps).
- Automate disclosures (e.g., APR calculations for transparency).

Other Features

30. Seasonal Loan Products

- Configure products with dynamic repayment schedules (e.g., higher repayments post-harvest).

31. API Integration for Rates

- Fetch real-time interest rates from central banks/financial markets.

32. Bundled Products

- Combine loans with insurance/other services (e.g., \"Trade Finance Package\").

33. Simulation Tools

- Borrowers can simulate loan costs before applying (what-if scenarios).

34. Product Performance Analytics

- Track uptake, default rates, profitability per product.
- Compare performance across regions/time periods.

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Loan Application & Approval

Required Features

35. Multi-Channel Application Submission

- Borrowers can apply via:

35.1. Web portal (full feature)

35.2. Mobile app (optimized for low-bandwidth)

35.3. USSD (for feature phones)

35.4. Other channels

36. Application Types

36.1. Individual Loans: Personal/consumer loans

36.2. Business Loans: SME/corporate financing

36.3. Group Loans: Joint liability groups (e.g., farmer cooperatives)

36.4. Emergency Loans: Fast-tracked processing

37. Smart Form Features

37.1. Auto-fill borrower details from existing CRM profiles

37.2. Dynamic forms that show/hide fields based on:

I. Loan type selected

II. Borrower category (individual vs. business)

III. Collateral requirements

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37.3. In-line validations (e.g., ID number format checks)

38. Document Management
    - Required documents per loan type:

Loan Type	Mandatory Documents
Individual	ID, proof of income, bank statements
Business	Certificate of incorporation, audited accounts
Group	Member IDs, group registration certificate

- File upload with:
    - 38.1. Format restrictions (PDF, JPG, PNG)
    - 38.2. Size limits (max 10MB per file)
    - 38.3. OCR for automatic data extraction (e.g., reading ID numbers)

39. Application Processing
    - 39.1. Auto-generated application ID (format: EDF-YYYYMMDD-XXX)
    - 39.2. System checks for:
    - 39.3. Duplicate applications
    - 39.4. Blacklisted borrowers
    - 39.5. Incomplete documentation

Risk scoring based on:

40. Credit bureau data
41. Existing loan exposure
42. Collateral coverage ratio

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43. Multi-Level Approval Workflows

    - Configurable approval chains:

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- Parallel approvals for cross-functional sign-offs (e.g., legal + credit)

Communication System

- Automated alerts via:

I. Email (with secure document links)

II. SMS (local language support)

III. In-app notifications

Reminder Schedule:

I. First notice: 5 days before due (SMS+email)

II. Second notice: Due date (WhatsApp+IVR call)

III. Third notice: 3 days overdue (registered letter)

Message Templates:

I. Local language support

II. Dynamic fields: (amountdue, duedate)

III. Payment link embedding

IV. Status update templates:

**\"Dear \[Name\], your application EDF-20240530-045 is now \[status\].
Next step: \[action\]\"**

Process Flow diagram description

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Collateral & Guarantor Management

Required Features

44. Collateral Registration

a. Types of Collateral Supported:

i. Real Estate (land, buildings)

ii. Vehicles (cars, machinery)

iii. Financial Instruments (cash deposits, stocks)

iv. Commodities (agricultural produce, inventory)

v. Insurance Policies

vi. Intellectual Property (patents, trademarks)

b. Collateral Details Capture:

i. Ownership documents (title deeds, registration certificates)

ii. Valuation reports (from approved appraisers)

iii. Insurance information (policy number, coverage, expiry)

iv. Location details (physical address + geotagging)

v. Photos/diagrams of collateral

45. Collateral Valuation

a. Automated value calculations:

i. Fixed assets: Use depreciation schedules

ii. Commodities: Link to market price APIs

iii. Cash deposits: Current account balance

b. Revaluation triggers:

i. Time-based (annual revaluation)

ii. Event-based (natural disasters, market crashes)

46. Collateral-Loan Linking

I. Multiple collateral types per loan

II. Cross-collateralization tracking

III. Dynamic Loan-to-Value (LTV) ratio monitoring:

![A close-up of a diagram Description automatically generated](media/image5.png){width="7.15625in" height="1.4232983377077866in"}

47. Guarantor Management

a. Guarantor Types:

i. Individual (personal guarantors)

ii. Corporate (company guarantees)

iii. Joint-and-Several (group guarantees)

b. Guarantor Onboarding:

i. KYC verification

ii. Financial capacity assessment

iii. Relationship to borrower (family, business partner)

48. Exposure Monitoring

a. Real-time dashboards showing:

i. Guarantor\'s total exposure

ii. Available guarantee capacity

iii. Concentration risks

b. Automatic alerts when:

i. Guarantor reaches limit

ii. Guarantor\'s credit score drops

49. Document Management

a. Secure storage for:

i. Collateral insurance certificates

ii. Guarantee agreements

iii. Notarized documents

b. Automated expiry reminders (insurance renewals)

50. Field Inspection Module

a. Schedule physical inspections

b. Mobile app for officers to:

i. Capture inspection photos

ii. Record GPS coordinates

iii. Submit condition reports

51. Collateral Release Process

I. Automatic release upon loan closure

II. Partial release approvals

III. Audit trail for all release actions

Process Flows For Collateral Registration

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Guarantor Management Process Flow

![A diagram of a financial assessment Description automatically generated](media/image7.png){width="2.65625in" height="4.07417760279965in"}

Validation Rules

52. Collateral Validation:

I. Minimum income coverage greater than or equal to the loan value

II. Property collateral must have clear title

53. Guarantor Validation:

I. Maximum 3 or more concurrent guarantees per individual

II. Corporate guarantors must be profitable for 2+ years

III. Guarantor net worth must exceed guarantee amount

54. Cross-Checking:

I. Prevent borrower and guarantor being same person

II. Flag related-party guarantees for extra scrutiny

Reporting

55. Collateral Reports:

I. Under-secured loans report

II. Insurance expiry calendar

III. Depreciation schedules

56. Guarantor Reports:

I. Guarantor concentration report

II. Guarantor call history

III. Exposure by industry sector

57. Regulatory Reports:

I. Collateral coverage ratios

II. Guarantee exposure limits

III. Stress test results

Security Features

58. Document Protection:

I. Watermarking of all certificates

II. Encryption at rest and in transit

59. Access Control:

I. Set a principle for collateral releases

II. Role-based access to valuation data

60. Audit Trails:

I. Full history of collateral modifications

II. Guarantee agreement version control

Repayment & Collections Management

Required Features

61. Multi-Channel Repayment Processing

- Supported Channels:

Channel Type	Processing Time	Fee Handling	Reconciliation Method
Bank Transfer	Real-time	Deduct from principal	Automated MT942
Mobile Money	Instant	Added to installment	API callbacks

Cash (Agent/Branch)	EOD batch	Separate fee payment	Manual verification
Payroll Deduction	Monthly cycle	Pre-tax deduction	Employer CSV upload

- Channel-Specific Features:
  - 61.1. Mobile money: USSD confirmation codes
  - 61.2. Bank transfers: Virtual account numbers per borrower
  - 61.3. Cash: GPS-tagged receipt generation

62. Dynamic Payment Allocation Engine
    - Payment Application Hierarchy:
    - 62.1. Outstanding penalties
    - 62.2. Accrued interest
    - 62.3. Principal amount
    - 62.4. Future installments (for overpayments)
    - Special Cases:
    - 62.5. Partial payments: Prorate across components
    - 62.6. Bullet payments: Trigger re-amortization
    - 62.7. Early repayments: Apply discount calculations
63. Late Payment Management
    - Penalty Calculation Matrix:

Days Late	Penalty Type	Calculation Basis

1-7	Fixed fee	2% of installment
8-30	Progressive interest	1.5x normal rate on overdue
31+	Compound penalties	Daily accrual on total arrears

- Grace Period Configuration:

  63.1. Product-level settings (0-15 days)

  63.2. Religious/holiday exceptions

  63.3. Disaster moratorium triggers

64. Restructuring & Recalculation

  - Supported Adjustments:

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- Auto-Recalculation Rules:

I. Interest capitalization options

II. Residual term vs. payment amount preferences

III. Regulatory ceiling validations

- Collections Workbench

  - Officer Tools:

  64.1. Delinquency segmentation (30/60/90+ days)

  64.2. Call scripts with compliance prompts

  64.3. Settlement negotiation logging

- Automated Actions:

64.4. Payment plan generation

64.5. Collateral realization triggers

64.6. Legal process initiation

- Receipting & Documentation
  - Auto-Generated Documents:

64.7. Payment receipts (QR-coded)

64.8. Annual interest certificates

64.9. Loan closure letters

Tax Handling(withholding tax, SACCO/NSSF exemptions)

Process Flows for Standard Repayment:

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Restructuring Process:

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Validation Rules

65. Payment Validation:

I. Minimum payment = 1% of outstanding or \$5 (whichever higher)

II. Duplicate payment detection

III. Currency conversion thresholds

66. Restructuring Checks:

I. Maximum 2 restructures per loan

II. Minimum 6 months between restructures

III. Board approval for \>25% principal reduction

67. Compliance Rules:

I. Usury law validations

II. Cooling-off period enforcement

III. Right-of-offset controls

Reporting Framework

68. Operational Reports:

I. Collection efficiency ratio

II. Channel performance analysis

III. Agent productivity metrics

69. Risk Reports:

I. Roll rate analysis

II. Provisioning calculations

III. Cure rate tracking

70. Regulatory Reports:

I. NPL reporting (CBK format)

II. Dormant account handling

III. Compliance Reports

Repayment & Collections

71. User (borrower) can make repayments via multiple channels (bank, mobile money).

72. System auto-calculates penalties for late payments.

73. System recalculates interest on restructuring/bullet payments.

74. System sends repayment reminders (5 days and 1 day before due date).

6. Debt Recovery

75. System flags delinquent loans based on repayment history.

76. Admin can issue demand notices and track recovery actions.

77. System prioritizes delinquent accounts for collection.

7. Reporting & Analytics

78. Admin can generate reports:

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79. Loan portfolio status (active, delinquent, repaid).

80. Aging analysis (PAR 30/60/90).

81. Expected Credit Loss (ECL) reports.

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82. System provides a dashboard with KPIs (disbursements, arrears, revenue).

8. User & Access Control

83. User (borrower) can register and log in (individuals and businesses).

84. Admin can manage user roles (loan officers, approvers, super admins).

85. System enforces role-based access control (e.g., only approvers can disburse loans).

Optional Features

1. Enhanced User Experience

86. User can reset password via email/SMS.

87. User can view loan statements (transaction history).

88. User can request loan restructuring via self-service portal.

2. Advanced Collateral Management

89. Geotagging of collateral (Google Maps integration).

90. Automated collateral value updates (e.g., commodity price fluctuations).

3. Additional Reporting

91. User can export reports to Excel/PDF.

92. Customizable dashboards (drag-and-drop widgets).

4. Integration & Automation

93. Credit Bureau API integration (pull/push credit reports).

94. Auto-sync with accounting systems (GL/expenses tracking).

5. Risk Management

95. Fraud detection alerts (unusual loan application patterns).

96. Insurance expiry alerts (for collateral).